Basic System Information

PATH Environment Variable

PATH is an environment variable that tells Windows where to search for executable files when you run a command.

When you type a command:

ipconfig

Windows checks each directory listed in PATH until it finds ipconfig.exe.

View PATH Variable

Command:

set

Look for:

Path=C:\Windows\system32;C:\Windows;...

```
C:\>set
ALLUSERSPROFILE=C:\ProgramData
[...]
LOGNAME=strategos
NUMBER_OF_PROCESSORS=2
OS=Windows_NT
Path=C:\Windows\system32;C:\Windows;C:\Windows\System32\Wbem;C:\
[...]
```

Checking Windows Version

Command:

ver

```
C:\>ver
Microsoft Windows [Version 10.0.17763.1821]
```

Purpose:

- Displays Windows OS version.
- Example:
  - Microsoft Windows [Version 10.0.17763.1821]

Getting Detailed System Information

Command:

systeminfo

Displays:

- Host Name
- OS Name
- OS Version
- Installed RAM
- Processor details
- Domain / Workgroup
- System uptime
- Network configuration

```
C:\>systeminfo

Host Name:                 WIN-SRV-2019
OS Name:                   Microsoft Windows Server 2019
OS Version:                10.0.17763 N/A Build 17763
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Standalone Server
OS Build Type:             Multiprocessor Free
[...]
```

Handling Long Output (Piping)

If output is too long:

Use pipe operator |

```
driverquery | more
```

- driverquery generates output
- | more displays it page by page
- Press Spacebar → Next page
- Press CTRL + C → Exit

This is called output redirection / piping.

Basic Utility Commands

help

- Displays help information.
- Example:
  - `help systeminfo`

cls

- Clears the screen.
- Command:

```
cls
```

Windows Networking Commands

Network Configuration

1. ipconfig

Displays basic IP configuration.

Command: `ipconfig`

```
Connection-specific DNS Suffix  . : eu-west-1.compute.internal
Link-local IPv6 Address . . . . . : fe80::90df:4861:ba40:f2a8%4
IPv4 Address. . . . . . . . . . . : 10.10.230.237
Subnet Mask . . . . . . . . . . . : 255.255.0.0
Default Gateway . . . . . . . . . : 10.10.0.1
```

Shows:

- IPv4 address
- IPv6 address
- Subnet mask
- Default gateway

2. Ipconfig /all

Displays detailed network configuration.

Command: `ipconfig /all`

Additional Information:

- MAC address (Physical Address)
- DHCP status
- DHCP server
- Lease time
- DNS servers
- Adapter description
- NetBIOS status

```
C:\>ipconfig /all

Ethernet adapter Ethernet 3:

   Connection-specific DNS Suffix  . : eu-west-1.compute.internal
   Description . . . . . . . . . . . : Amazon Elastic Network Adapter
   Physical Address. . . . . . . . . : 02-B7-DF-1D-0D-99
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::90df:4861:ba40:f2a8%4(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.10.230.237(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Lease Obtained. . . . . . . . . . : Wednesday, May 1, 2024 2:38:05 PM
   Lease Expires . . . . . . . . . . : Wednesday, May 1, 2024 4:08:07 PM
   Default Gateway . . . . . . . . . : 10.10.0.1
   DHCP Server . . . . . . . . . . . : 10.10.0.1
   DHCPv6 IAID . . . . . . . . . . . : 134353458
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-27-E3-D1-2B-0E-F8-30-D0-72-3F
   DNS Servers . . . . . . . . . . . : 10.0.0.2
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

Network Troubleshooting

1. ping

Checks connectivity using ICMP Echo Request/Reply.

Command: `ping target_name`

Output Shows:

- IP resolution
- Round Trip Time (RTT)
- Packet loss
- TTL value
- Key metrics:

➔ 0% loss → reachable
➔ High latency → possible congestion
➔ 100% loss → unreachable or ICMP blocked

2. tracert

Traces route to a target.

Command: `tracert example.com`

How It Works

- Sends packets with increasing TTL (Time-to-Live)
- Routers reply when TTL expires
- Reveals hop-by-hop path

```
C:\>tracert example.com

Tracing route to example.com [93.184.215.14]
over a maximum of 30 hops:

  1    59 ms    32 ms    42 ms  ec2-3-248-240-3.eu-west-1.compute.amazonaws.com [3.248.240.3]
  2     *        *        *     Request timed out.
  3     *        *        *     Request timed out.
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6     *        *        *     Request timed out.
  7     *        *        *     Request timed out.
  8     *        *        *     Request timed out.
  9    <1 ms    13 ms    <1 ms  100.100.2.56
 10    15 ms    11 ms    11 ms  ae-42.a03.londen12.uk.bb.gin.ntt.net [131.103.117.104]
 11    17 ms    11 ms    12 ms  ae-14.r20.londen12.uk.bb.gin.ntt.net [129.250.3.248]
 12    81 ms    80 ms    80 ms  ae-7.r20.nwrknj03.us.bb.gin.ntt.net [129.250.6.147]
 13    83 ms    83 ms    86 ms  ae-0.a02.nycmny17.us.bb.gin.ntt.net [129.250.3.9]
 14    79 ms    79 ms    96 ms  ce-0-3-0.a02.nycmny17.us.ce.gin.ntt.net [128.241.1.14]
 15    81 ms    86 ms    79 ms  ae-67.core1.nyd.edgecastcdn.net [152.195.68.135]
 16    78 ms    78 ms    78 ms  93.184.215.14

Trace complete.
```

Output Shows:

- Each router on path
- Latency per hop
- Where timeout occurs

DNS Lookup

1. nslookup

Resolves domain name to IP address.

Command: `nslookup example.com`

Uses default DNS server.

Specify DNS Server: `nslookup example.com 1.1.1.1`

- Uses a specific DNS server.
- 1.1.1.1 belongs to: Cloudflare

Output Shows:

- DNS server used
- IP address of target
- Whether answer is authoritative

```
C:\>nslookup example.com
Server:  ip-10-0-0-2.eu-west-1.compute.internal
Address:  10.0.0.2

Non-authoritative answer:
Name:    example.com
Addresses:  2606:2800:21f:cb07:6820:80da:af6b:8b2c
          93.184.215.14

C:>nslookup example.com 1.1.1.1
Server:  one.one.one.one
Address:  1.1.1.1

Non-authoritative answer:
Name:    example.com
Addresses:  2606:2800:21f:cb07:6820:80da:af6b:8b2c
          93.184.215.14
```

Viewing Network Connections

1. netstat

Basic Command: `netstat`

Shows: Established connections only

```
C:\>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    10.10.230.237:22       ip-10-11-81-126:53486  ESTABLISHED
```

2. netstat -abon

Flag	Meaning
-a	All connections & listening ports
-b	Shows executable name
-o	Shows PID
-n	Numerical format (no DNS resolution)

Command: `netstat -abon`

```
C:\>netstat -abon

Active Connections

  Proto  Local Address          Foreign Address        State           PID
  TCP    0.0.0.0:22             0.0.0.0:0              LISTENING       2116
 [sshd.exe]
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING       820
  RpcSs
 [svchost.exe]
[...]
  TCP    0.0.0.0:49669          0.0.0.0:0              LISTENING       2036
 [spoolsv.exe]
  TCP    0.0.0.0:49670          0.0.0.0:0              LISTENING       584
 Can not obtain ownership information
  TCP    0.0.0.0:49686          0.0.0.0:0              LISTENING       592
 [lsass.exe]
  TCP    10.10.230.237:22       10.11.81.126:53486     ESTABLISHED     2116
 [sshd.exe]
 [...]
```

What You Can Identify

- Listening ports
- Associated service/executable
- Process ID
- Active connections
- Remote IPs

Example:

➔ 0.0.0.0:22 LISTENING [sshd.exe]
➔ Port 22 → SSH

➔ 0.0.0.0:135 LISTENING
➔ RpcSs
➔ Port 135 → RPC

➔ 0.0.0.0:3389 LISTENING
➔ TermService
➔ Port 3389 → RDP

Working With Directories

Check Current Directory

1. cd

Example output:
C:\Users\strategos

List Directory Contents

1. dir

Shows:

- Files
- Folders
- File size
- Date modified
- Free disk space

```
C:\Users\strategos>cd
C:\Users\strategos

C:\Users\strategos>dir
 Volume in drive C has no label.
 Volume Serial Number is A8A4-C362

 Directory of C:\Users\strategos

05/01/2024  02:40 PM    <DIR>          .
05/01/2024  02:40 PM    <DIR>          ..
11/14/2018  06:56 AM    <DIR>          Desktop
05/01/2024  02:40 PM    <DIR>          Documents
09/15/2018  07:19 AM    <DIR>          Downloads
09/15/2018  07:19 AM    <DIR>          Favorites
09/15/2018  07:19 AM    <DIR>          Links
09/15/2018  07:19 AM    <DIR>          Music
09/15/2018  07:19 AM    <DIR>          Pictures
09/15/2018  07:19 AM    <DIR>          Saved Games
09/15/2018  07:19 AM    <DIR>          Videos
               0 File(s)              0 bytes
              11 Dir(s)  14,984,953,856 bytes free
```

Command		Purpose
dir /a	→	Show hidden & system files
dir /s	→	Show files in current directory + all subdirectories

Display Folder Tree Structure

1. tree
Shows visual hierarchy of directories.

```
C:\Users\strategos>tree
Folder PATH listing
Volume serial number is A8A4-C362
C:.
├───Desktop
├───Documents
├───Downloads
├───Favorites
├───Links
├───Music
├───Pictures
├───Saved Games
└───Videos
```

Useful for:

- Understanding structure quickly
- Reporting directory layout
- CTF enumeration

Change Directory

1. cd folder1 (change directory to folder1)
2. cd .. (go back one level)

Managing directory and files

Create directory: mkdir foldername
Delete directory: rmdir foldername

View file:

1. type filename.txt (small file)
2. more filename.txt (large file)

Copy Files: copy source destination
Move files: move source destination
Delete files:

1. del filename
2. erase filename

Wildcards

Wildcard symbol: * (means “all”)

- Example:
  - copy *.md C:\Markdown
- Copies all .md files to C:\Markdown.

- Other examples:
  - del *.txt
- Deletes all .txt files in the current directory.

Windows Process Management

Listing Running Processes

Basic Command:

tasklist

Displays:

- Image Name (process name)
- PID (Process ID)
- Session Name
- Session #
- Memory Usage

```
C:\>tasklist

Image Name                     PID Session Name        Session#    Mem Usage
========================= ======== ================ =========== ============
System Idle Process              0 Services                   0          8 K
System                           4 Services                   0         88 K
Registry                        84 Services                   0     50,700 K
smss.exe                       276 Services                   0      1,132 K
csrss.exe                      372 Services                   0      5,264 K
wininit.exe                    448 Services                   0      6,892 K
csrss.exe                      456 Console                    1      5,028 K
winlogon.exe                   516 Console                    1     11,144 K
services.exe                   584 Services                   0      7,492 K
lsass.exe                      592 Services                   0     16,108 K
svchost.exe                    704 Services                   0     23,432 K
fontdrvhost.exe                736 Console                    1      4,256 K
[...]
```

Filtering Processes

The output of the tasklist can be very long. Use filters.

1. View Help

tasklist /?

Shows available filtering options.

```
C:\>tasklist /FI "imagename eq sshd.exe"

Image Name                     PID Session Name        Session#    Mem Usage
========================= ======== ================ =========== ============
sshd.exe                      2116 Services                   0      6,992 K
sshd.exe                      2712 Services                   0      7,668 K
sshd.exe                      4752 Services                   0      7,372 K
```

2. Filter by Image Name

tasklist /FI "imagename eq sshd.exe"

Explanation:

1. /FI → Filter
2. imagename → Field to filter
3. eq → Equals
4. sshd.exe → Target process

Understanding PID (Process ID)

Each running process has a unique PID.

Why PID matters:

- Used to terminate process
- Used in netstat -abon
- Used in forensic investigations
- Used in privilege escalation labs

Killing a Process

1. Kill by PID

taskkill /PID 4567
(Terminates the process with PID 4567)

2. Force Kill

taskkill /PID 4567 /F
(/F → Force termination)

3. Kill by Image Name

taskkill /IM sshd.exe
(Kills all processes with that name)

Force version:

taskkill /IM sshd.exe /F